

Splits, dividends and the return you actually measure

Daily learning note | 1 October 2026 | Q008

Sidiq review status: Not yet reviewed.

Research question and hypothesis

Can an explicit share-and-cash calculation distinguish raw price changes from holding-period returns across a split and dividend? The hypothesis is that valuing the same opening holding at both endpoints removes mechanical split effects and counts each dividend entitlement once. Q006 established when research decisions occur; Q008 now establishes a return arithmetic contract before membership, features and targets. This is the next eligible non-Wednesday task after C002.

Beginner explanation and definitions

A share split changes the size of each share. Imagine exchanging one large slice for two smaller slices of the same cake. Seeing each new slice priced at half the original price does not mean your holding lost half its value. A dividend also changes what you own: some wealth may move from the share price into a cash entitlement. We must count both pieces in consistent units.

Raw price: unadjusted price per share at that endpoint. **Split ratio s :** ending shares per opening share, so a 2-for-1 split has $s=2$. **Reverse split:** consolidation of shares, such as $s=0.2$ for 1-for-5. **Cash entitlement C :** dividend amount due per opening share. **Capital return:** change in the value of the shares, excluding cash. **Income return:** dividend entitlement divided by opening value. **Total holding-period return:** combined change in share value and entitlement over one interval.

Ex-dividend date: date governing whether a purchase carries the next dividend entitlement under the applicable rules. **Payment date:** when the distribution is paid. An entitlement may exist before cash settles; today's model values that receivable at par and does not make it available for trading.

Data, metrics and predeclared protocol

All eight examples are synthetic, undated single holding intervals. There is no market-data sample, fit, security selection, backtest or empirical performance claim. Prices are raw, positive and in one currency; the holder is assumed entitled to the specified distribution. No within-interval reinvestment.

Before implementation, acceptance required split-only and dividend hand examples, a reverse split, a combined event exposing unit errors, invalid-input rejection, independent holdings arithmetic and repeatable output. Metrics are component/total returns, hand-case passes and detected wrong controls. A $1e-12$ absolute/relative comparison tolerance checks finite calculations; it is not a universal accuracy bound. Unknown action inputs must fail rather than silently become no action. Full project checks and a verified teaching PDF are required before publication.

Mathematics from one opening share

Let P_0 be the opening price, P_1 the closing price, s the ending shares per opening share and C the cash entitlement per opening share. Opening wealth $W_0 = P_0$. Closing equity $E_1 = s \cdot P_1$ and closing wealth $W_1 = E_1 + C$. Return is the change in wealth divided by opening wealth: $R = (s \cdot P_1 + C) / P_0 - 1$.

The raw price return is $P_1 / P_0 - 1$. The capital return is $s \cdot P_1 / P_0 - 1$. The income contribution is C / P_0 . Thus total return equals capital return plus income contribution, up to numeric rounding. For any positive opening quantity q , both W_0 and W_1 multiply by q , which cancels in the return ratio.

Hand-calculated examples

Scenario	Opening to closing holding	Raw price change	Total return
Ordinary gain	1*100 to 1*110	10%	10%
Ordinary loss	1*100 to 1*90	-10%	-10%
2-for-1 split	1*100 to 2*50	-50%	0%
1-for-5 reverse split	1*20 to 0.2*100	400%	0%
Dividend offsets price fall	1*100 to 1*98 + 2	-2%	0%
Dividend with gain	1*100 to 1*103 + 2	3%	5%
Split and dividend	1*100 to 2*49 + 2	-51%	0%
Split, dividend and gain	1*100 to 2*54 + 2	-46%	10%

In the combined zero-return example, ending equity is 98 and the entitlement is 2, so total wealth remains 100. Capital return is -2%, income contributes +2%, and total return is 0%. A dividend quoted as 1 per ending share must become 2 per opening share before entering this function.

Three deliberately wrong controls

Ignoring the split in the split-only example reports $50/100 - 1 = -50\%$. Inverting the split reports $(50/2)/100 - 1 = -75\%$. Both are false losses caused by inconsistent share units. The correct result is zero.

Multiplying the already normalized cash by the split again gives $2 \cdot (49 + 2) / 100 - 1 = +2\%$ in the combined example. That counts 4 units of cash when the holder is entitled to 2. Passing a post-split quote of 1 directly as C instead creates a -1% result. Parameter names help, but cannot authenticate units.

Scope of the mathematics

Pure split neutrality is an accounting example, not a forecast that a real price stays fixed through an event. The examples deliberately choose offsetting prices to expose errors. Actual market movement can create gains or losses alongside a split or dividend. The reverse-split example permits fractional shares; real cash-in-lieu arrangements need their own evidence and are outside this calculation.

Implementation and reproduction

src/quant_research/data/corporate_actions.py adds action_return and an immutable ActionReturn result. The required keyword inputs are shares_per_old_share and cash_per_old_share. Values 1 and 0 explicitly assert known absence of an action. Missing arguments, None, strings, booleans, nonfinite values, nonpositive prices/ratio and negative cash are rejected. Unknown actions are not filled with defaults.

Nonfinite intermediates and complete underflow of positive quantities raise ValueError. Scalar prices cannot reveal adjusted prices, mixed currencies or incorrect events; those are upstream obligations.

scripts/corporate_action_experiment.py records all eight inputs/results and three wrong controls in experiments/corporate_actions/results.json. The companion README describes assumptions and sources; tests/test_corporate_actions.py supplies independent hand values and invalid controls. No dependency was added. Run make corporate-action-experiment, make check and make note NOTE_DATE=2026-10-01.

Observed results and checks

Check	Observed result
New Python tests	52 pass; all eight hand component examples agree
Independent ledger oracle	162 exact-Fraction comparisons over 54 cases and three currency scales
Wrong calculations	Ignored/inverted split and double-counted cash all detected
Reproduction	Two subprocess runs match archived JSON bytes
Complete Python suite	415 tests pass
Quality checks	Ruff lint/format passes; strict mypy passes on 18 source files
C++ regression	Fresh warning-clean C++17 build; 66 checks and golden output pass
Environment and provenance	Dependency consistency passes; four original Q003 hashes verify

The independent oracle starts from 15 shares, values the closing shares and cash using exact rational arithmetic, then compares with the floating-point function. Scaling all money amounts by 0.01, 1 and 100 leaves the expected return unchanged. This tests units and wealth reconciliation without claiming exhaustive input coverage. No coverage percentage or speed improvement was measured.

Primary sources consulted

SEC Investor.gov, Stock Split: explains the relation between share count, price and ownership. <https://www.investor.gov/introduction-investing/investing-basics/glossary/stock-split>

SEC Investor.gov, Ex-Dividend Dates: distinguishes entitlement dates, payment timing and special cases. <https://www.investor.gov/introduction-investing/investing-basics/glossary/ex-dividend-dates-when-are-you-entitled-stock-and>

Consulted 1 October 2026. The equations follow our synthetic contract; no vendor equivalence is asserted.

Limitations and interview explanation

This is one-interval arithmetic, not an event processor or portfolio ledger. It assumes raw prices, known entitlement and a common currency. Applying a split again to split-adjusted prices, or adding a dividend to a total-return-adjusted series, can double-count actions. The scalar interface cannot detect that mistake. Upstream metadata and licensed sample checks remain necessary.

No announcement, ex-date, payment-date or event-order validation is implemented here. Recognizing a receivable at par does not imply it has settled. Chaining returns with reinvestment requires a separate policy for timing and financing. Fees, taxes, FX, dividend reinvestment, rights, mergers, spin-offs, terminal proceeds and fractional-share rounding are excluded. A missing terminal price never justifies zero recovery; zero endpoint prices are rejected under this narrow scope.

Floating-point rounding remains even when all checks pass. Extreme intermediate values may be rejected although a rescaled calculation would fit. Small effects can round away; the range guards are not a complete precision guarantee. The hand examples establish arithmetic correctness, not economic realism.

For an interview, begin with a holding rather than a price chart. State the share basis and dividend units, derive closing wealth, then distinguish raw price, capital and total return. Explain why unknown action evidence stays unresolved and why entitlement is different from settled cash. These repository results are material to study, not a claim that personal understanding has already been demonstrated.

Next work and necessary input

Q008 is complete. Q009a historical membership intervals is Ready for October 2; Q009b will separately address terminal-event evidence. C003 stays in the Wednesday October 7 slot. No input is needed for this synthetic continuation. Empirical research remains blocked on Q004a lawful entitlement/sample evidence and Q005b integration. Existing licensed access or a source/budget preference is needed before acquisition; credentials should not be sent or committed.

Sidiq review remains Not yet reviewed. Publication is authorised separately. Daily runs remain at 08:00 Europe/London, including daylight-saving changes. Stop new research after 31 December 2026 and request the next programme scope.

Check your understanding

1. Why does a 2-for-1 split from 100 to 50 not imply a 50% loss for the holder?
2. After a 2-for-1 split, a dividend is quoted as 1 per new share. What is C?
3. Why is an ex-date entitlement not necessarily cash available to buy more shares?
4. Can this function safely identify already adjusted prices or missing terminal proceeds?

Suggested answers

1. The holder now owns two shares: $2 \times 50 = 100$. The raw prices refer to different share units.
2. $C=2$ per opening share. Multiply the quoted 1 by the two ending shares once, then use that amount.
3. Payment may occur later. Today's receivable is valued at par; settlement and financing are not modeled.
4. No. Price basis must be verified upstream. Terminal-event evidence is separate, and zero/missing endpoint prices are rejected rather than interpreted as a loss or recovery amount.